



Mathematics Part 1 — Cbse Annual Exam — 80 Marks

ANSWER KEY — FOR GRADING

Subject: Mathematics Part 1 | Class: Class 9

Total Marks: 80 | Time Allowed: 3h | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 20 questions. Each carries 1 mark(s).		
1	How many parts do the coordinate axes divide the plane into? (A) 2 (B) 6 (C) 4 (D) 8 Answer: Correct Option: (C) ► Why? C. The axes divide the plane into four parts, called quadrants. [Ch 1: Orienting Yourself: The Use of Coordinates]	1
2	Using the identity $(a + b)^2 = a^2 + 2ab + b^2$, what is the expansion of $(5x + 2y)^2$? (A) $25x^2 + 10xy + 4y^2$ (B) $5x^2 + 20xy + 2y^2$ (C) $5x^2 + 10xy + 2y^2$ (D) $25x^2 + 20xy + 4y^2$ Answer: Correct Option: (D) ► Why? D. Substituting $a = 5x$ and $b = 2y$ into the identity yields $(5x)^2 + 2(5x)(2y) + (2y)^2 = 25x^2 + 20xy + 4y^2$. [Ch 4: Exploring Algebraic Identities]	1



Q.NO.	QUESTIONS	MARKS
3	Two coins are tossed at the same time. What is the probability of getting at least one head? (A) $\frac{1}{4}$ (B) $\frac{3}{4}$ (C) $\frac{1}{2}$ (D) 1 Answer: Correct Option: (B) ► Why? B. The sample space is {HH, HT, TH, TT}. The event 'at least one head' includes {HH, HT, TH}, which is 3 out of 4 outcomes, giving a probability of $\frac{3}{4}$. [Ch 7: The Mathematics of Maybe: Introduction to Probability]	1
4	Assertion (A): 471 is not a term of the sequence given by the formula $s_n = 5n - 2$. Reason (R): For a number to be a term of the sequence, the value of n must be a natural number. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (A) ► Why? Both A and R are true, and R is the correct explanation of A. Solving $5n - 2 = 471$ gives $n = 94.6$, which is not a natural number, so 471 is not a term of the sequence. The requirement that n must be a natural number, as stated in R, is the exact reason why 471 is not a term. [Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]	1
5	By the Baudhāyana–Pythagoras Theorem, what is the length of the diagonal of a square where each side is exactly 1 unit long? (A) $\sqrt{2}$ (B) 2 (C) 1 (D) $\sqrt{3}$ Answer: Correct Option: (A) ► Why? A. The length of the diagonal is $\sqrt{2}$ because $1^2 + 1^2 = d^2$ yields $d^2 = 2$, so $d = \sqrt{2}$. [Ch 3: The World of Numbers]	1
6	A line drawn from the centre of a circle to the midpoint of a chord is always: (A) Perpendicular to the chord (B) Parallel to the chord (C) Equal to the chord (D) A diameter of the circle Answer: Correct Option: (A)	1



Q.NO.	QUESTIONS	MARKS
	► Why? A. A line drawn from the centre to the midpoint of a chord is perpendicular to that chord. [Ch 5: I'm Up and Down, and Round and Round]	
7	What value must the fourth side d of a 4-gon take for Brahmagupta's formula to reduce to Heron's formula? (A) 1 (B) a (C) -1 (D) 0 Answer: Correct Option: (D) ► Why? D. When $d = 0$, the 4-gon collapses into a triangle, and Brahmagupta's formula reduces to Heron's formula. [Ch 6: Measuring Space: Perimeter and Area]	1
8	What is the simplified fractional form of the pure repeating decimal $0.4\bar{5}$? (A) $45/100$ (B) $9/20$ (C) $5/11$ (D) $15/90$ Answer: Correct Option: (C) ► Why? C. Let $x = 0.4\bar{5}$. Multiplying by 100 gives $100x = 45.4\bar{5}$. Subtracting the equations yields $99x = 45$, so $x = 45/99 = 5/11$. [Ch 3: The World of Numbers]	1
9	The area of a sector of a circle with radius r that subtends an angle θ° at the centre is: (A) $\pi r^2 \times \theta/180$ (B) $\pi r^2 \times \theta/360$ (C) $2\pi r \times \theta/360$ (D) $\pi r^2 \times \theta/90$ Answer: Correct Option: (B) ► Why? B. The area of a sector is $\pi r^2 \times \theta^\circ/360^\circ$, using the rotational symmetry of the circle. [Ch 6: Measuring Space: Perimeter and Area]	1
10	In the linear relationship $y = ax + b$, what does the constant b represent? (A) The slope of the line (B) The x-intercept (C) The degree of the polynomial (D) The y-intercept, which is the distance from the origin where the line cuts the y-axis	1



Q.NO.	QUESTIONS	MARKS
	Answer: Correct Option: (D) ► Why? D. In the equation $y = ax + b$, the constant b is called the y-intercept, which is the distance from the origin where the line cuts the y-axis. [Ch 2: Introduction to Linear Polynomials]	
11	What does probability measure? (A) Physical quantities like length and volume (B) The likelihood of events (C) The exact outcome of an experiment (D) The speed of moving objects Answer: Correct Option: (B) ► Why? B. Probability is a type of measurement used to measure the likelihood of events, expressing how confident or certain we are that a particular event will occur. [Ch 7: The Mathematics of Maybe: Introduction to Probability]	1
12	Assertion (A): When triangle ADM is reflected in the y-axis to form triangle A'D'M', the length A'D' equals the length AD. Reason (R): The distance between two points is given by $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, and it makes no difference whether $(x_2 - x_1)$ and $(y_2 - y_1)$ are positive or negative. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (A) ► Why? A. The assertion is true as the source confirms that reflection preserved the lengths of the sides, with $A'D' = 5$ units equal to $AD = 5$ units. The reason is true and correctly explains this: reflecting in the y-axis changes the signs of the x-coordinates, which changes the signs of $(x_2 - x_1)$ terms, but since the distance formula squares these differences, the sign does not matter and the distance remains the same. [Ch 1: Orienting Yourself: The Use of Coordinates]	1
13	Which Greek mathematician described the latitudes and longitudes of thousands of locations, including 'Ozine' (Ujjayinī)? (A) Hipparchus (B) Āryabhaṭa (C) Ptolemy (D) Baudhāyana Answer: Correct Option: (C) ► Why? C. The Greek mathematician Ptolemy (c. 150 BCE) later described the latitudes and longitudes of thousands of locations.	1



Q.NO.	QUESTIONS	MARKS
	<i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	
14	Assertion (A): The Virahānka-Fibonacci sequence is an arithmetic progression. Reason (R): In an arithmetic progression, the difference between consecutive terms is constant. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (D) ► Why? A is false but R is true. The Virahānka-Fibonacci sequence does not have a constant difference between its consecutive terms, so it is not an arithmetic progression. However, R correctly states the defining property of an arithmetic progression. <i>[Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]</i>	1
15	If two opposite angles of a quadrilateral add up to 180°, what can be concluded about its vertices? (A) They form a parallelogram (B) They are collinear (C) They lie on a circle (are concyclic) (D) They are equidistant from the centre Answer: Correct Option: (C) ► Why? C. If two opposite angles of a quadrilateral add up to 180°, then the vertices of the quadrilateral lie on a circle, i.e., they are concyclic. <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	1
16	In the equation $y = ax + b$, what does a represent? (A) Slope of the line (B) y-intercept of the line (C) x-intercept of the line (D) Origin Answer: Correct Option: (A) ► Why? A. In the equation $y = ax + b$, a represents the slope of the line and b represents the y-intercept. <i>[Ch 2: Introduction to Linear Polynomials]</i>	1
17	Which of the following identities represents the factorization of $x^3 - y^3$? (A) $(x - y)(x^2 - xy + y^2)$ (B) $(x - y)(x^2 + xy + y^2)$ (C) $(x + y)(x^2 - xy + y^2)$ (D) $(x + y)(x^2 + xy + y^2)$	1



Q.NO.	QUESTIONS	MARKS
	Answer: Correct Option: (B) ► Why? B. The identity for the difference of cubes is $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$. [Ch 4: Exploring Algebraic Identities]	
18	Assertion (A): In a random experiment like tossing a coin, you cannot predict exactly what will happen. Reason (R): Randomness refers to a situation where you may know all the possible outcomes, but you cannot say which one will definitely occur. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (A) ► Why? A. Both A and R are true, and R is the correct explanation of A. The assertion is true because the outcome of a coin toss cannot be predicted in advance. The reason correctly defines randomness, which explains why exact prediction is impossible even when all possible outcomes are known. [Ch 7: The Mathematics of Maybe: Introduction to Probability]	1
19	The nth term of the odd number sequence is given by $u_n = 2n - 1$. Which position does the number 137 occupy in this sequence? (A) 67 (B) 69 (C) 68 (D) 70 Answer: Correct Option: (B) ► Why? B. Setting $u_n = 137$ gives $2n - 1 = 137$, which simplifies to $2n = 138$ or $n = 69$. [Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]	1
20	Assertion (A): The product $(x-y)(x^2+xy+y^2)$ simplifies to $x^3 + y^3$. Reason (R): When $(x-y)$ is multiplied by (x^2+xy+y^2) using the distributive property, the terms x^2y and xy^2 cancel out. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (D) ► Why? D. The assertion is false because the product simplifies to $x^3 - y^3$, not $x^3 + y^3$. The reason is true as the intermediate terms x^2y and xy^2 do indeed cancel out during the distribution process. [Ch 4: Exploring Algebraic Identities]	1



Q.NO.	QUESTIONS	MARKS
Section B Attempt 5 out of 7.		
21	Why is π classified as an irrational number? Answer: π is classified as an irrational number because it cannot be written as a ratio of two integers. Its decimal digits go on forever without any repeating pattern, unlike rational numbers which have rhythmic decimal expansions. ► Why? This question tests the understanding of the property of irrational numbers applied to π, contrasting it with the properties of rational numbers. [Ch 6: Measuring Space: Perimeter and Area]	2
	OR Describe the construction steps to locate $\sqrt{2}$ on the number line using a unit square's diagonal. Answer: Measure OA = 1 unit on the number line and draw a perpendicular at A. Mark point B such that AB = 1 unit, making OB = $\sqrt{2}$ units. With O as center and OB as radius, draw an arc intersecting the number line at P, representing $\sqrt{2}$. ► Why? This tests the geometric construction method described in the text, relying on the Baudhāyana-Pythagoras Theorem to create the length $\sqrt{2}$ and transferring it via a compass. [Ch 3: The World of Numbers]	
22	Find the value of 119^2 by expressing 119 as $100 + 10 + 9$ and applying a suitable identity. Answer: Using $(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$, we get $100^2 + 10^2 + 9^2 + 2(100)(10) + 2(100)(9) + 2(10)(9) = 10000 + 100 + 81 + 2000 + 1800 + 180 = 14161$. ► Why? This tests the ability to apply the three-variable square identity to compute a large square, following the specific example provided. [Ch 4: Exploring Algebraic Identities]	2
	OR A chord of length 24 cm is drawn in a circle of diameter 26 cm. Find the distance from the centre of the circle to the chord. Answer: The radius is 13 cm. The perpendicular from the centre bisects the chord, forming a right triangle with half the chord (12 cm) and the radius (13 cm). The distance from the centre is $\sqrt{(13^2 - 12^2)} = \sqrt{(169 - 144)} = 5$ cm. ► Why? The perpendicular from the centre to the chord bisects it, creating a right triangle where the radius is the hypotenuse, allowing the use of the Pythagorean theorem. [Ch 5: I'm Up and Down, and Round and Round]	
23	In a survey of 50 students, 15 students said they liked football. The number of students who like football is 15, and the ____ (frequency/relative frequency) is ____ (fill in the fraction or decimal).	2



Q.NO.	QUESTIONS	MARKS
	Answer: relative frequency, 0.3 (or 15/50) ► Why? According to the provided context, the ratio of the number of times an event occurs to the total number of trials is called the relative frequency (e.g., relative frequency of rolling a 4 is $8/50 = 0.16$). Here, the number of students who like football is 15 out of a total of 50 students. Thus, the relative frequency is calculated as $15/50$, which simplifies to 0.3. ► Step-by-step: 1. Step 1: Identify the term for the ratio of observed occurrences to total trials, which is relative frequency. 2. Step 2: Calculate the relative frequency using the formula: Relative frequency = (Number of students who like football) / (Total number of students) = $15 / 50 = 0.3$. [Ch 7: The Mathematics of Maybe: Introduction to Probability]	
24	What role did Ujjayinī play in the ancient world regarding coordinates? Answer: Ujjayinī was described as the point marking the central longitude meridian from which all other locations were measured. This was noted at least as early as the 4th century BCE in the early Siddhāntas. ► Why? This assesses the student's knowledge of the historical application of coordinates to the Earth's surface, specifically the significance of Ujjayinī. [Ch 1: Orienting Yourself: The Use of Coordinates]	2
25	An auto-rickshaw fare is ₹25 for the initial 2 km and increases by ₹15 per km thereafter. Write the simplified linear expression for the total fare for n km, where $n \geq 2$. Answer: The total fare for n km is $25 + 15(n - 2) = 15n - 5$. This expression is valid for distances where $n \geq 2$. It combines the fixed initial fare with the variable cost for the remaining distance. ► Why? The expression accounts for the fixed charge for the first 2 km and the variable charge for the remaining $(n - 2)$ km, simplifying to a linear form. [Ch 2: Introduction to Linear Polynomials]	2
Section C Attempt 5 out of 7.		
26	The number $0.\bar{9}$ (which means 0.99999 ...) is a rational number. Using algebra (let $x = 0.\bar{9}$, multiply by 10, and subtract), explain why $0.\bar{9}$ is exactly equal to 1.	3



Q.NO.	QUESTIONS	MARKS
	Answer: Let $x = 0.99999\dots$. This means the number has an infinitely repeating block of 9s. Next, we multiply by 10 to shift the decimal point one place to the right. This gives $10x = 9.99999\dots$ Now, we subtract the first equation from the second: $10x - x = 9.99999\dots - 0.99999\dots$ $9x = 9$ Dividing both sides by 9, we get $x = 1$. Therefore, $0.99999\dots$ is exactly equal to 1. This demonstrates the non-uniqueness of decimal representations, where a terminating decimal like $1.000\dots$ has an alternative form with repeating 9s. ► Why? The question asks to prove $0.999\dots = 1$ algebraically. By setting $x = 0.999\dots$, multiplying by 10 gives $10x = 9.999\dots$. Subtracting the original equation yields $9x = 9$, so $x = 1$. The context also mentions that rational numbers can have two decimal forms, showing non-uniqueness where any terminating decimal has an alternative with repeating 9s. ► Step-by-step: - Step 1: Let $x = 0.99999\dots$ - Step 2: Multiply by 10 to get $10x = 9.99999\dots$ - Step 3: Subtract the two equations: $10x - x = 9.99999\dots - 0.99999\dots$ - Step 4: Simplify to get $9x = 9$ - Step 5: Solve for x to get $x = 1$ [Ch 3: The World of Numbers]	
	OR Determine the AP whose third term is 16 and whose 7th term exceeds the 5th term by 12. Answer: 4, 10, 16, 22, 28, ... ► Why? The problem requires finding the first term 'a' and common difference 'd' of an AP using the given conditions. The formula for the nth term of an AP, $t_n = a + (n-1)d$, is applied to the 3rd, 5th, and 7th terms to form equations. Solving these equations yields the values of a and d, which are then used to write the complete AP. ► Step-by-step: - Step 1: Let the first term of the AP be 'a' and the common difference be 'd'. The nth term is given by $t_n = a + (n-1)d$. - Step 2: Given the 3rd term is 16, so $a + 2d = 16$. - Step 3: Given the 7th term exceeds the 5th term by 12, so $(a + 6d) - (a + 4d) = 12$. Simplifying this gives $2d = 12$, so $d = 6$. - Step 4: Substitute $d = 6$ into $a + 2d = 16$ to get $a + 12 = 16$, which gives $a = 4$. - Step 5: Using $a = 4$ and $d = 6$, the required AP is 4, 10, 16, 22, 28, ... [Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]	
27	Solve the previous question using the Baudhāyana–Pythagoras theorem.	3



Q.NO.	QUESTIONS	MARKS
	Answer: Given: $CE \perp AB$, $CH \perp FG$, and $CE = CH$. Since the perpendicular from the centre bisects the chord, E and H are the midpoints of AB and FG, respectively. Applying the Baudhāyana-Pythagoras theorem to right triangles $\triangle CEA$ and $\triangle CHF$, we get: $CA^2 = CE^2 + AE^2$ and $CF^2 = CH^2 + FH^2$. Since CA and CF are radii of the same circle, $CA = CF$. Therefore, $CE^2 + AE^2 = CH^2 + FH^2$. We are given that $CE = CH$, which means $CE^2 = CH^2$. Substituting this into our equation, we get $AE^2 = FH^2$, which implies $AE = FH$. Because E and H are midpoints, $AB = 2AE$ and $FG = 2FH$. Since $AE = FH$, we can conclude that $AB = FG$. Hence, chords equidistant from the centre have equal length. ► Why? The question asks to prove Theorem 7 (chords equidistant from the centre have equal length) using the Baudhāyana-Pythagoras theorem. The context provides the logic for Theorem 8 which uses this theorem to compare distances of unequal chords. By applying the theorem to the right triangles formed by the radius, perpendicular distance, and half the chord, we can equate the squares of the radii. Since the distances from the centre are equal, their squares cancel out, leaving the squares of the half-chords equal, thus proving the chords are equal. ► Step-by-step: 1. Step 1: Apply the Baudhāyana-Pythagoras theorem to right $\triangle CEA$ and $\triangle CHF$: $CA^2 = CE^2 + AE^2$ and $CF^2 = CH^2 + FH^2$. 2. Step 2: Equate the expressions for CA^2 and CF^2 since $CA = CF$ (radii), resulting in $CE^2 + AE^2 = CH^2 + FH^2$. 3. Step 3: Use the given condition $CE = CH$ to cancel CE^2 and CH^2, leaving $AE^2 = FH^2$, which means $AE = FH$. 4. Step 4: Since E and H are midpoints, $AB = 2AE$ and $FG = 2FH$. Because $AE = FH$, it follows that $AB = FG$. [Ch 5: I'm Up and Down, and Round and Round]	
	OR Find the value of the quadratic polynomial $7s^2 - 4s + 6$ when $s = -3$. Answer: Given: The quadratic polynomial is $7s^2 - 4s + 6$, and the value of $s = -3$. Formula: Substitute the value of the variable into the expression and apply the correct order of operations. Substitution: Value = $7(-3)^2 - 4(-3) + 6$ Calculation: = $7(9) + 12 + 6$ = $63 + 12 + 6$ = $75 + 6$ = 81 ∴ The value of the quadratic polynomial at $s = -3$ is 81.	



Q.NO.	QUESTIONS	MARKS
	► Why? Substituting a negative value into a quadratic polynomial requires careful application of exponent rules and sign conventions. ► Step-by-step: - Step 1: Substitute $s = -3$ into the expression $7s^2 - 4s + 6$. - Step 2: Evaluate the exponent $(-3)^2 = 9$, then multiply by 7 to get 63. - Step 3: Multiply -4 by -3 to get +12. - Step 4: Add the terms $63 + 12 + 6$ to get the final answer 81. [Ch 2: Introduction to Linear Polynomials]	
28	Define a segment of a circle. A chord of a circle of radius 10 cm subtends 90° at the centre. Find the area of the corresponding minor sector. (Use $\pi \approx 3.14$) Answer: A segment of a circle is the region bounded by an arc of the circle and the chord joining the endpoints of the arc. Given: Radius (r) = 10 cm, Angle (θ) = 90°. Formula: Area of sector = $\pi r^2 \times \theta/360^\circ$. Substitution: Area = $3.14 \times 10^2 \times 90/360$ Area = $3.14 \times 100 \times 1/4$ Calculation: Area = $3.14 \times 25 = 78.5 \text{ cm}^2$ ∴ The area of the minor sector is 78.5 cm^2. ► Why? A segment is defined geometrically by an arc and its chord. The minor sector's area is found by applying the sector area formula with the given central angle. ► Step-by-step: - Step 1: Define a segment as the region bounded by an arc and its chord. - Step 2: Identify the given values for the sector: $r = 10 \text{ cm}$ and $\theta = 90^\circ$. - Step 3: Apply the sector area formula: $\pi r^2 \times \theta/360^\circ$. - Step 4: Substitute values and calculate the area as 78.5 cm^2. [Ch 6: Measuring Space: Perimeter and Area]	3
29	What is a random experiment in probability? Give one example and explain why it qualifies as random.	3



Q.NO.	QUESTIONS	MARKS
	Answer: Definition: A random experiment is something you can repeat, where every time you do it, the result might be different and you cannot know the outcome in advance. Example: Tossing a fair coin is a random experiment. Explanation: It qualifies as random because you can repeat the toss multiple times, and although you know the possible outcomes are Heads or Tails, you cannot predict exactly which one will come up on a single toss. The outcome is unpredictable. Answer: ∴ A random experiment is repeatable with unpredictable outcomes, like tossing a coin. ► Why? Randomness relies on the inability to predict a specific outcome in advance despite knowing all possible results, making repeated trials vary in their results. ► Step-by-step: 1. Step 1: Define a random experiment as a repeatable action with unpredictable outcomes. 2. Step 2: Provide an example like tossing a fair coin. 3. Step 3: Explain that while possible outcomes (Heads, Tails) are known, the exact result of a single toss cannot be predicted. [Ch 7: The Mathematics of Maybe: Introduction to Probability]	
30	State the general formula used to find the distance between two points (x_1, y_1) and (x_2, y_2) in the 2-D plane. Why does it not matter whether $(x_2 - x_1)$ and $(y_2 - y_1)$ are positive or negative? Answer: The general formula for the distance between two points (x_1, y_1) and (x_2, y_2) is: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ This formula is derived using the Baudhāyana–Pythagoras Theorem, where $(x_2 - x_1)$ and $(y_2 - y_1)$ represent the shifts along the two axes. It makes no difference whether these differences are positive or negative because squaring a negative value yields a positive result. We are simply measuring the shifts along the two axes, and the total length depends on the magnitudes of these shifts, not their direction. ► Why? The answer states the distance formula from the text and explains that squaring eliminates the impact of negative signs, as we are only measuring the magnitude of shifts. ► Step-by-step: 1. Step 1: Write the distance formula: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ 2. Step 2: Recognize that the formula involves squaring the differences of the coordinates. 3. Step 3: Conclude that since squaring yields positive values, the signs of the differences do not matter as we are just measuring shifts along the axes. [Ch 1: Orienting Yourself: The Use of Coordinates]	3



Q.NO.	QUESTIONS	MARKS
Section D Attempt 4 out of 6.		
31	Suppose a plant has height 1.75 feet and it grows by 0.5 feet each month. (i) Find the height after 7 months. (ii) Make a table of values for t varying from 0 to 10 months and show how the height, h, increases every month. (iii) Find an expression that relates h and t, and explain why it represents linear growth.	5



Q.NO.	QUESTIONS	MARKS																								
	Answer: (i) (i) Height after 7 months: The initial height of the plant is 1.75 feet, and it grows by 0.5 feet each month. Given: Initial height = 1.75 feet, Growth rate = 0.5 feet/month, Time = 7 months Formula: Height = Initial height + (Growth rate × Time) Substitution: Height = 1.75 + (0.5 × 7) Calculation: Height = 1.75 + 3.5 = 5.25 feet ∴ The height after 7 months is 5.25 feet. (ii) (ii) Table of values: The height increases by a constant amount of 0.5 feet every month. The table of values for time t (in months) and height h (in feet) is as follows: <table><tr><td>t</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>h</td><td>1.75</td><td>2.25</td><td>2.75</td><td>3.25</td><td>3.75</td><td>4.25</td><td>4.75</td><td>5.25</td><td>5.75</td><td>6.25</td><td>6.75</td></tr></table> As shown in the table, the height increases by 0.5 feet every month. This constant increase over equal intervals is a key characteristic of the data provided. (iii) (iii) Expression and explanation: The expression that relates height h and time t is: $h = 1.75 + 0.5t$ This represents linear growth because, as the value of t increases by a fixed number (one month), the value of the height h increases by a fixed number (0.5 feet). A linear pattern where a quantity increases by a constant amount over equal intervals represents linear growth. ∴ The expression is $h = 1.75 + 0.5t$, representing linear growth.	t	0	1	2	3	4	5	6	7	8	9	10	h	1.75	2.25	2.75	3.25	3.75	4.25	4.75	5.25	5.75	6.25	6.75	
t	0	1	2	3	4	5	6	7	8	9	10															
h	1.75	2.25	2.75	3.25	3.75	4.25	4.75	5.25	5.75	6.25	6.75															



Q.NO.	QUESTIONS	MARKS
	► Why? The initial height is 1.75 feet, and the plant grows 0.5 feet per month. (i) After 7 months, the height is: $1.75 + (0.5 \times 7) = 5.25 \text{ feet}$ (ii) The table shows the height at each month from 0 to 10, increasing by 0.5 feet each time. (iii) The expression is derived from the initial value plus the rate of growth times the time: $h = 1.75 + 0.5t$ It represents linear growth because the quantity increases by a constant amount over equal intervals, as defined in the text. ► Step-by-step: 1. Step 1: Calculate height after 7 months: $\text{Initial height} + (\text{Growth rate} \times \text{Time}) = 1.75 + (0.5 \times 7) = 1.75 + 3.5 = 5.25 \text{ feet}$ 2. Step 2: Create table for $t = 0$ to 10. For each month, add 0.5 to the previous height. $t = 0: 1.75; t = 1: 2.25; t = 2: 2.75; \dots; t = 10: 6.75.$ 3. Step 3: Formulate the expression: $h = \text{Initial height} + (\text{Growth rate} \times t) = 1.75 + 0.5t$ 4. Step 4: Explain linear growth: The height increases by a constant amount (0.5 feet) over equal intervals (1 month), which defines linear growth. [Ch 2: Introduction to Linear Polynomials]	
	OR A city has two main roads which cross each other at the centre of the city. These two roads are along the North-South (N-S) direction and East-West (E-W) direction. All the other streets of the city run parallel to these roads and are 200 m apart. There are 10 streets in each direction. (i) Using 1 cm = 200 m, draw a model of the city in your notebook. Represent the roads/streets by single lines. (ii) There are street intersections in the model. Each street intersection is formed by two streets—one running in the N-S direction and another in the E-W direction. Each street intersection is referred to in the following manner: If the second street running in the N-S direction and 5th street in the E-W direction meet at some crossing, then we call this street intersection (2, 5). Using this convention, find: (a) how many street intersections can be referred to as (4, 3). (b) how many street intersections can be referred to as (3, 4). Answer: (i) Draw two perpendicular lines intersecting at the center (Origin). Using 1 cm = 200 m, draw 5 parallel lines at 1 cm intervals on both sides of the N-S road and 5 parallel lines at 1 cm intervals on both sides of the E-W road.	



Q.NO.	QUESTIONS	MARKS
	► Why? The city model represents a 2-D Cartesian coordinate system where the main roads act as the x-axis and y-axis, crossing at the origin (0,0). Because distances to the right and upwards are positive, and distances to the left and downwards are negative, a specific street number (like the 4th street) exists on both the positive and negative sides of the origin. Therefore, any intersection (a, b) will have four possible locations in the grid: (a, b), (a, -b), (-a, b), (-a, -b) ► Step-by-step: 1. Step 1: Recognize the city model as a 2-D Cartesian coordinate system with the center crossing as the origin (0,0). 2. Step 2: Understand that N-S streets represent x-coordinates (positive to the East, negative to the West) and E-W streets represent y-coordinates (positive to the North, negative to the South). 3. Step 3: For intersection (4, 3), the 4th N-S street can be $x = 4$ or $x = -4$. The 3rd E-W street can be $y = 3$ or $y = -3$. The four possible intersections are: (4, 3), (4, -3), (-4, 3), (-4, -3) 4. Step 4: For intersection (3, 4), the 3rd N-S street can be $x = 3$ or $x = -3$. The 4th E-W street can be $y = 4$ or $y = -4$. The four possible intersections are: (3, 4), (3, -4), (-3, 4), (-3, -4) [Ch 1: Orienting Yourself: The Use of Coordinates]	
32	Prove that the opposite angles of a cyclic quadrilateral add up to 180°. Explain how the angles subtended by arcs at the centre and at the circumference are used in this proof.	5



Q.NO.	QUESTIONS	MARKS
	Answer: A cyclic quadrilateral is a quadrilateral whose all four vertices lie on a circle. The fundamental property of such a quadrilateral is that its opposite angles are supplementary, i.e., they add up to 180°. Consider a cyclic quadrilateral ABCD inscribed in a circle with centre O. Draw the radii OB and OD. This divides the circle into two arcs: arc BCD and arc BAD. The angle $\angle BAD$ is subtended by arc BCD at point A on the circle. Since A lies outside arc BCD, the angle that arc BCD subtends at the centre O is the reflex angle BOD (going from OB to OD along C). By the property that the angle at the circumference is half the angle at the centre: $\angle BAD = \frac{1}{2} (\text{reflex } \angle BOD)$ Similarly, the angle $\angle BCD$ is subtended by arc BAD at point C on the circle. The angle that arc BAD subtends at the centre is the angle BOD (going from OB to OD via A). So: $\angle BCD = \frac{1}{2} (\angle BOD)$ Adding both angles: $\angle BAD + \angle BCD = \frac{1}{2} (\text{reflex } \angle BOD) + \frac{1}{2} (\angle BOD) = \frac{1}{2} (\text{reflex } \angle BOD + \angle BOD)$ The reflex angle BOD and the angle BOD together form the complete angle at the centre O, which equals 360°. Therefore: $\angle BAD + \angle BCD = \frac{1}{2} \times 360^\circ = 180^\circ$ By the same reasoning applied to the other pair, $\angle ABC + \angle ADC = 180^\circ$. This confirms that in any cyclic quadrilateral, each pair of opposite angles sums to 180°. Diagram description: Draw a circle with centre O. Mark four points A, B, C, D on the circle in order. Join OB and OD. Show the reflex angle BOD (dotted arc from OB to OD going through C) and the angle BOD (from OB to OD going through A). Label $\angle BAD$ and $\angle BCD$. ► Why? The proof uses the arc-angle relationship — the angle subtended by an arc at the centre is twice the angle it subtends at any point on the remaining part of the circle — to show that the two opposite angles together cover half the complete rotation at the centre. ► Step-by-step: Step 1: Define a cyclic quadrilateral and state what needs to be proved. Step 2: Identify the two arcs (arc BCD and arc BAD) and the corresponding angles at the centre (reflex $\angle BOD$ and $\angle BOD$). Step 3: Apply the property that angle at circumference = $\frac{1}{2} \times$ angle at centre for both opposite angles. Step 4: Add the two expressions and use the fact that reflex angle + angle = complete angle = 360°. Step 5: Conclude that the sum equals $\frac{1}{2} \times 360^\circ = 180^\circ$ and describe the supporting diagram.	



Q.NO.	QUESTIONS	MARKS
	<i>[Ch 5: I'm Up and Down, and Round and Round]</i>	
	OR State the identity for $(a + b + c)^2$ and describe its geometric interpretation using a square of side $(a + b + c)$. Use this identity to find the value of 119^2 and to expand $(3x - 2y + 4z)^2$.	



Q.NO.	QUESTIONS	MARKS
	Answer: The identity for the square of the sum of three numbers is: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ Geometric Interpretation: A square of side $(a + b + c)$ is drawn. The side is divided into three segments of lengths a, b, and c. Lines parallel to the sides are drawn at these division points, creating a grid of 9 regions inside the square. These regions consist of: • Three squares with areas a^2, b^2, and c^2. • Six rectangles — two each with areas ab, bc, and ca. The total area of the big square equals the sum of all these regions, giving the identity: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ Finding 119^2: Given: $119 = 100 + 10 + 9$, so $a = 100$, $b = 10$, $c = 9$. Formula: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ Substitution: $119^2 = (100 + 10 + 9)^2$ $= 100^2 + 10^2 + 9^2 + 2(100)(10) + 2(100)(9) + 2(10)(9)$ $= 10000 + 100 + 81 + 2000 + 1800 + 180$ $= 14161$ $\therefore 119^2 = 14161$ Expanding $(3x - 2y + 4z)^2$: Given: $a = 3x$, $b = -2y$, $c = 4z$ Formula: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ Substitution: $(3x - 2y + 4z)^2 = (3x)^2 + (-2y)^2 + (4z)^2 + 2(3x)(-2y) + 2(-2y)(4z) + 2(4z)(3x)$ $= 9x^2 + 4y^2 + 16z^2 - 12xy - 16yz + 24xz$	



Q.NO.	QUESTIONS	MARKS
	$\therefore (3x - 2y + 4z)^2 = 9x^2 + 4y^2 + 16z^2 - 12xy - 16yz + 24xz$ ► Why? The identity $(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ arises from partitioning a square of side $(a+b+c)$ into 3 smaller squares and 6 rectangles, and it simplifies both numerical squaring and algebraic expansion. ► Step-by-step: Step 1: State the identity $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$. Step 2: Describe the geometric model — a square of side $(a+b+c)$ divided into 3 squares and 6 rectangles, and label each region with its area. Step 3: Apply the identity to find 119^2 by expressing 119 as $100 + 10 + 9$ and substituting into the formula. Step 4: Expand $(3x - 2y + 4z)^2$ by identifying $a = 3x$, $b = -2y$, $c = 4z$ and carefully substituting while tracking signs. Step 5: Simplify all terms and state the final expanded expression. [Ch 4: Exploring Algebraic Identities]	
33	Find 5 rational numbers between $\frac{2}{5}$ and $\frac{3}{5}$. Show that the rational number $\frac{(a+b)}{2}$ always lies between any two rational numbers a and b . Using this result, explain what is meant by the density of rational numbers on the number line.	5



Q.NO.	QUESTIONS	MARKS
	Answer: Given: Two rational numbers $a = \frac{2}{5}$ and $b = \frac{3}{5}$ Finding 5 rational numbers between $\frac{2}{5}$ and $\frac{3}{5}$: Multiply numerator and denominator by 6 to increase the gap between numerators: $\frac{2}{5} = \frac{12}{30}$ and $\frac{3}{5} = \frac{18}{30}$ The integers between 12 and 18 are: 13, 14, 15, 16, 17 $\therefore$ Five rational numbers between $\frac{2}{5}$ and $\frac{3}{5}$ are: $\frac{13}{30}, \frac{14}{30}, \frac{15}{30}, \frac{16}{30}, \frac{17}{30}$ Proof that $(a+b)/2$ lies between a and b: If $a < b$, then: Adding a to both sides: $a + a < a + b$, so $2a < a + b$ Adding b to both sides: $a + b < b + b$, so $a + b < 2b$ Therefore: $2a < a + b < 2b$ Dividing by 2: $a < (a+b)/2 < b$ $\therefore (a+b)/2$ lies strictly between a and b. Density of rational numbers means that a rational number always exists between any two other rational numbers. Since $(a+b)/2$ is rational when a and b are rational, and it lies between them, we can repeatedly apply this to find infinitely many rational numbers between any two given ones. This shows rational numbers are infinitely dense on the number line — between any two points, no matter how close, there is always another rational number. ► Why? Rational numbers are dense: between any two rational numbers, their average $(a+b)/2$ is another rational number, and this process can be repeated infinitely, filling the number line densely. ► Step-by-step: 1. Step 1: Express both fractions with a common denominator large enough to create a gap of more than 5 between numerators. 2. Step 2: List the integers between the two numerators to get the desired rational numbers. 3. Step 3: Prove $(a+b)/2$ lies between a and b using inequalities: if $a < b$, then $2a < a+b < 2b$, so $a < (a+b)/2 < b$. 4. Step 4: Explain density — since $(a+b)/2$ is always rational and lies between a and b, rational numbers always exist between any two rational numbers, making them dense on the number line.	



Q.NO.	QUESTIONS	MARKS
	[Ch 3: The World of Numbers]	
34	A triangle has side lengths of 3 units, 4 units, and 5 units. Calculate its area using Heron's formula. Then, verify the area using the 'half base times height' formula, justifying why the triangle is right-angled. Answer: Given: Triangle with sides a = 3 units, b = 4 units, c = 5 units. Formula (Heron's): Semi-perimeter, $s = 1/2(a + b + c)$ Area = $\sqrt{s(s - a)(s - b)(s - c)}$ Calculation (Heron's): $s = 1/2(3 + 4 + 5) = 6$ units Area = $\sqrt{6 \times (6 - 3) \times (6 - 4) \times (6 - 5)}$ Area = $\sqrt{6 \times 3 \times 2 \times 1} = \sqrt{36} = 6$ sq. units Verification: Since $3^2 + 4^2 = 5^2$, by the converse of the Baudhāyana-Pythagoras theorem, the triangle is right-angled. This means the two shorter sides act as the base and the height. Formula (Base $\times$ Height): Area = $1/2 \times \text{base} \times \text{height}$ Calculation (Base $\times$ Height): For the right-angled triangle, base = 3 units, height = 4 units. Area = $1/2(3 \times 4) = 6$ sq. units $\therefore$ The area is 6 sq. units by both formulas. ► Why? Heron's formula calculates area purely from side lengths, while the base-times-height formula is verified here by recognizing the 3-4-5 triangle as right-angled using the converse of the Baudhāyana-Pythagoras theorem. ► Step-by-step: - Step 1: Calculate the semi-perimeter s using the given side lengths. - Step 2: Substitute the values of s, a, b, and c into Heron's formula to find the area. - Step 3: Identify the triangle as right-angled using the converse of the Baudhāyana-Pythagoras theorem. - Step 4: Apply the $1/2 \times \text{base} \times \text{height}$ formula using the two shorter sides as base and height. - Step 5: Conclude that both formulas yield the exact same area. [Ch 6: Measuring Space: Perimeter and Area]	5



Q.NO.	QUESTIONS	MARKS
Section E Attempt 3 out of 4.		
35	Case Study: City Map Coordinates A student is creating a 2-D map of a city center, inspired by how early Arabic maps used the Ujjayinī meridian as a zero-longitude reference. On the map, the main intersection is chosen as the point of intersection of the axes, called the origin. The horizontal road is the x-axis and the vertical road is the y-axis. A cafe is located 3 units to the right and 2 units upwards from the origin. A library is located 4 units to the left and 5 units upwards from the origin. A park is located 2 units to the left and 3 units downwards from the origin. The student marks these locations using coordinates. (i) What are the coordinates of the origin? (ii) The park is located 2 units to the left and 3 units downwards from the origin. In which quadrant does the park lie? (iii) The cafe is at (3, 2). A shop is located at (7, 2). Find the distance between the cafe and the shop. (iv) A bookstore is located on the y-axis, 6 units upwards from the origin, and a florist is located on the x-axis, 6 units to the right of the origin. Are the coordinates of the bookstore and the florist the same? Explain. Answer: (i) (0, 0) (ii) The third quadrant (iii) The distance between points (x1, y) and (x2, y) is $x_2 - x_1$. Here, distance = $7 - 3 = 4$ units. (iv) No, the bookstore is at (0, 6) and the florist is at (6, 0). Since $0 \neq 6$, $(0, 6) \neq (6, 0)$. ► Why? i) (0, 0) ii) The third quadrant iii) The distance between points (x1, y) and (x2, y) is $x_2 - x_1$. Here, distance = $7 - 3 = 4$ units. iv) No, the bookstore is at (0, 6) and the florist is at (6, 0). Since $0 \neq 6$, $(0, 6) \neq (6, 0)$. <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	5
	OR Case Study: Algebraic Expressions and Terms A shop has sealed boxes of different colours for sale. Red boxes have 4 pens each and blue boxes have 5 pencils each. If a customer buys x red boxes and y blue boxes and gets 3 extra pens free, the total quantity of pens and pencils is given by the algebraic expression $4x + 5y + 3$. In this expression, $4x$, $5y$ and 3 are terms, x and y are variables, the numbers 4 and 5 are the coefficients of x and y respectively, and 3 is a constant. (i) What are the variables used in the expression $4x + 5y + 3$? (ii) What is the coefficient of y in the given expression? (iii) Identify the terms and the constant in the expression $4x + 5y + 3$. (iv) If a customer buys 2 red boxes and 1 blue box, what is the total quantity of pens and pencils they get?	



Q.NO.	QUESTIONS	MARKS
	Answer: (i) The variables used in the expression $4x + 5y + 3$ are x and y. (ii) The coefficient of y in the given expression is 5. (iii) The terms in the expression are $4x$, $5y$, and 3. The constant is 3. (iv) Substituting $x = 2$ and $y = 1$ in $4x + 5y + 3$, we get $4(2) + 5(1) + 3 = 8 + 5 + 3 = 16$. ► Why? i) The variables used in the expression $4x + 5y + 3$ are x and y. ii) The coefficient of y in the given expression is 5. iii) The terms in the expression are $4x$, $5y$, and 3. The constant is 3. iv) Substituting $x = 2$ and $y = 1$ in $4x + 5y + 3$, we get $4(2) + 5(1) + 3 = 8 + 5 + 3 = 16$. [Ch 2: Introduction to Linear Polynomials]	
36	Case Study: Finding Squares Using Identities A student was preparing for a mental math competition. She learned that identities simplify calculations significantly. To find 29^2, she wrote $(30 - 1)^2$ and used the identity $(a - b)^2 = a^2 - 2ab + b^2$ to get $900 - 60 + 1 = 841$. To find 119^2, she expressed it as $(100 + 10 + 9)^2$ and applied the identity $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$. She also learned that Śhrīdharāchārya in 750 CE proposed the identity $a^2 = (a + b)(a - b) + b^2$, which she used to find $55^2 = (55 + 5)(55 - 5) + 5^2 = 60 \times 50 + 25 = 3025$. (i) Using the identity $(a - b)^2 = a^2 - 2ab + b^2$, find the value of 79^2 by expressing 79 as $(80 - 1)$. (ii) To calculate 97^2, a student writes it as $(100 - 3)^2$. Which identity should she apply, and what will be the expanded form? (iii) Using the identity $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$, find the value of 117^2 by expressing 117 as $(100 + 10 + 7)$. (iv) Using Śhrīdharāchārya's method $a^2 = (a + b)(a - b) + b^2$, calculate 35^2 by choosing $b = 5$. Explain why $b = 5$ is a suitable choice. Answer: (i) 6241 (ii) The identity $(a - b)^2 = a^2 - 2ab + b^2$ should be applied. The expanded form is $100^2 - 2(100)(3) + 3^2$. (iii) $117^2 = 100^2 + 10^2 + 7^2 + 2(100)(10) + 2(100)(7) + 2(10)(7) = 10000 + 100 + 49 + 2000 + 1400 + 140 = 13689$ (iv) $35^2 = (35 + 5)(35 - 5) + 5^2 = 40 \times 30 + 25 = 1200 + 25 = 1225$. $b = 5$ is suitable because $35 + 5 = 40$ and $35 - 5 = 30$, both being round numbers that make the multiplication easy to compute mentally. ► Why? i) 6241 ii) The identity $(a - b)^2 = a^2 - 2ab + b^2$ should be applied. The expanded form is $100^2 - 2(100)(3) + 3^2$. iii) $117^2 = 100^2 + 10^2 + 7^2 + 2(100)(10) + 2(100)(7) + 2(10)(7) = 10000 + 100 + 49 + 2000 + 1400 + 140 = 13689$ iv) $35^2 = (35 + 5)(35 - 5) + 5^2 = 40 \times 30 + 25 = 1200 + 25 = 1225$. $b = 5$ is suitable because $35 + 5 = 40$ and $35 - 5 = 30$, both being round numbers that make the multiplication easy to compute mentally. [Ch 4: Exploring Algebraic Identities]	5



Q.NO.	QUESTIONS	MARKS
37	Case Study: Cyclic Quadrilateral Park A city park is shaped like a cyclic quadrilateral ABCD, with a circular walking path forming its boundary. The angles at the corners are carefully measured to ensure the design is accurate. The angle at corner A is 75 degrees, and the angle at corner B is 110 degrees. The park designer uses the properties of cyclic quadrilaterals to verify the layout. One such property states that opposite angles of a cyclic quadrilateral add up to 180 degrees. The converse property is also considered for future designs. (i) What is the measure of angle C? (ii) What is the measure of angle D? (iii) If angle A was $(2x + 10)$ degrees and angle C was $(3x - 20)$ degrees, find the value of x. (iv) State the converse property that determines if a quadrilateral is cyclic based on its opposite angles. Answer: (i) 105 degrees (ii) 70 degrees (iii) 38 (iv) If two opposite angles of a quadrilateral add up to 180 degrees, then the vertices of the quadrilateral lie on a circle (they are concyclic). ► Why? i) 105 degrees ii) 70 degrees iii) 38 iv) If two opposite angles of a quadrilateral add up to 180 degrees, then the vertices of the quadrilateral lie on a circle (they are concyclic). [Ch 5: I'm Up and Down, and Round and Round]	5

